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Regenerative business strategies: A database and typology to inspire business experimentation towards sustainability

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A R T I C L E I N F O

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A B S T R A C T

In a time marked by growing environmental and societal challenges, where multiple planetary boundaries are being crossed, the way we conduct business practices needs to be rethought. Regenerative business models offer a path towards a more responsible future by incorporating strategies that go beyond net zero and focus on actively restoring the natural and social systems they operate in, creating a more holistic net positive impact. To date, much of the research on this topic is conceptual and there is a need for more empirical insights on how regenerative strategies can be implemented. This study uses purposive sampling to conduct a comprehensive practice review of 84 regenerative business cases from 15 sectors, to build the novel Regenerative Business Case Database. A typology of six archetypal regenerative business strategies was also derived, including: (1) regenerative leadership, (2) nature regeneration, (3) social regeneration, (4) responsible sourcing, (5) human health & wellbeing focus, and (6) employee level focus. The predominant sectors with regenerative innovations were food, consumer goods and fashion. Many companies were found to be employing multiple strategies concurrently, while collaborating with local organisations. While some may argue that regenerative business strategies cannot be successful in a competitive market, the results contain examples of stable businesses pursuing regenerative strategies dating back to the 1870's. The aim of this study was to build a database and typology of successful regenerative business cases and strategies that can serve as inspiration for further business model experimentation. Future research could explore how innovations towards such business models develop, and what regulatory frameworks allow them to flourish.

1. Introduction

The need to mitigate and adapt to climate change is more significant than ever before. Approximately 17 % of the Amazon rainforest has been lost in the last 50 years due to deforestation and land use change, leading to vital biodiversity loss (Barredo, 2021). A 69 % decline in global wildlife populations has been observed since 1970 due to habitat destruction, climate change and pollution (WWF, 2022). The UN's Food and Agriculture Organisation estimates that 25 % of the Earth's land is highly degraded as a result of unsustainable agricultural practices, deforestation, and urbanisation (FAO, 2011). Further, a recent study by Richardson et al. (2023) documented that most planetary boundaries including biosphere integrity, land system change, biogeochemical flows are now being surpassed. These changes exacerbate the impacts of climate change that are already being felt globally, but more so for the 1.4 billion people who live in highly vulnerable areas, and whose livelihoods and food security are at risk (IPCC, 2022).

The developing concept of a circular economy has presented a promising new societal and production paradigm to combat these problems (Ghisellini et al., 2016). It can be described as a series of resource strategies that aim to close, narrow and slow resource loops (Bocken et al., 2016). Companies have increasingly set net circularity goals, as they try to incorporate these circular economy strategies and keep abreast with regulations (Bocken and Konietzko, 2022, 2023). Yet, research has suggested that the interpretation of circular economy by businesses in practice mainly emphasises closing and narrowing of loops, the strategies where often companies may also save cost while saving resources (Allwood, 2014; Ritala et al., 2018). However, while sound and much needed, these strategies can only go so far in creating a sustainable economy, as they mainly focus on harm reduction. Recent

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work which reported a global reduction in resource circularity from 9.1 % in 2018 to 7.2 % in 2023 due to increasing resource consumption, has suggested that a future circular economy must emphasise reductions, regeneration, and redistribution to counter this negative trend (Circle Economy, 2024). Reductions in total consumption of resources can be enabled through practicing sufficiency, and increasing product lifetimes (Bocken et al., 2022; Kaddoura et al., 2019). Redistribution is needed to transition towards a long-term flourishing circular society that is rooted in environmental and social prosperity, rather than a mere circular flow of materials (Bocken and Short, 2021; Jaeger-Erben et al., 2021). Finally, regeneration is needed, given the damage that has been done to the planet already (Persson et al., 2022; Richardson et al., 2023).

There is a need to go a step further than 'net zero', and actively regenerate the various spaces the economy operates in (Bocken and Short, 2021; Hoveskog et al., 2018). To regenerate in this context means to build stronger and more wholesome ecosystems and societies, using circular business practices (Sanford, 2017). The circular economy in its current interpretations by business and policy makers unfortunately does not effectively integrate elements of biodiversity regeneration, nature-inclusive development and social justice and resilience yet (Bosschaert, 2022). Regeneration is the much needed concept and practice of the hour on the levels of nature, society and economy (Muñoz and Branzei, 2021; Slawinski et al., 2021; Vlasov, 2021). From a business perspective, incorporating regenerative practices might lead to significant potential benefits in future risk management and building resilient supply chains (Hahn and Tampe, 2021; Rhodes, 2015).

Past research has investigated what it means to pursue regenerative business model strategies (Hahn and Tampe, 2021; Konietzko et al., 2023). There has also been some emerging literature that has cited some regenerative business cases (e.g., Gualandris et al., 2024; Hahn and Tampe, 2021; Konietzko et al., 2023) including some well-known company examples (e.g. Patagonia appointing nature as the main shareholder of its business, Ben & Jerry's fair and just trade model, Timberland's environmental stewardship for employees). But a comprehensive search, evaluation and analysis of regenerative business cases and patterns is still lacking. While there is growing business interest (Liu, 2024), and a practical need for regeneration and net positive strategies (Hawken, 2021; Polman and Winston, 2021). Not much is known about how regenerative business models are realized in practice, how they relate to their social and ecological environment, and how firms can approach embedding such strategies in their business models. While former research has developed typologies for sustainable and circular business models (Bocken et al., 2014; Lüdeke-Freund et al., 2019), such a typology does not exist for regenerative business. Therefore, building on emerging societal and environmental issues, and needs also expressed in popular work (Polman and Winston, 2021), we address the need for a more comprehensive understanding of regenerative business cases.

The research questions guiding this study were: *What are the different types of regenerative business models that exist in practice? What types of strategies are employed by businesses in order to be regenerative?*

To this end, this study conducts a comprehensive practice review to develop a regenerative business strategy typology and database from 84 regenerative business cases. The aim being to create an easily accessible database of real business case examples, that show regenerative business strategies that are being implemented in practice. This could incentivise and inspire other businesses to experiment with and adopt these business models that go beyond just minimising harm to nature, and instead actively focus on regenerating it.

In the following, Section 2 gives the background of this study, Section 3 describes the method followed, Section 4 describes the results, Section 5 gives the discussion, and Section 6 gives the conclusion.

2. Background

The following sections first describe the origins of the regeneration concept, then second its relevance in the context of regenerative business models, and lastly the greenwashing risks related to it.

2.1. The origins of the regeneration concept

Regeneration essentially refers to the ability of a system to remake or renew itself continuously. The term has its origins in biology and natural sciences, relating to the ability of cells, organisms and ecosystems to renew themselves (Carlson, 2011; Stinner et al., 1997). It has also been used in the field of conservation biology, as a proposition to combat biodiversity loss (Young, 2000). As a process it is essential to biological systems and describes their "*capacity to bring into existence again*" (Muñoz and Branzei, 2021, p. 509). The use of the term regeneration has become increasingly common in fields such as agriculture, design, conservation, tourism and built environment, where it is frequently used to describe processes that lead to the restoration or revitalisation of the living organisms, ecosystems or societal structures that they operate in (Konietzko et al., 2023). Given the damage done to the planet's natural systems and biodiversity through overexploitation by linear degenerative economies (Richardson et al., 2023), regeneration has also started to attract increasing attention in the context of creating better organisational practices (Muñoz and Branzei, 2021).

In the context of organisation and business, regeneration is envisioned to rebuild and strengthen ecosystems, making them more resilient to external forces (Carlson, 2011). Past research on regeneration in an organisational context started with early conceptualisations, for e.g., by Hardman (2009) who stressed the need to redefine leadership within organisations and pursue a new kind of regenerative leadership within organisations that is non-charismatic, and more purpose driven. Robinson and Cole (2015) gave the first conceptualisation of regenerative sustainability as a "net-positive approach" to sustainability, that would use business activities as a driver of positive change. The first mention of regeneration explicitly in the context of business models was in "The Regenerative Business" book by Carol Sanford (Sanford, 2017). Sanford (2017) described a regenerative business as one that serves both people and the planet, through improving systemic health and by using collaborative networks to transform the business ecosystems they operate in. This was followed by Dias (2018) who coined regenerative development, noting that regeneration can help build more resilient human and natural networks. The last few years have seen increasing amounts of attention on regeneration in the context of business, supply chains and the circular economy (Caldera et al., 2018; Gualandris et al., 2024; Howard et al., 2019; Morseletto, 2020; Muñoz and Branzei, 2021; van Hille et al., 2021). More recently, Hawken (2021) and Polman and Winston (2021) have repopularised the concept through books on ending the climate crisis through regeneration, and on 'net positive' as the future of sustainable business strategies, respectively.

As a result, the term regeneration is now finding new application within organisational sciences, where it is used to promote natural ecosystems and human societies that are wholesome and thriving (Slawinski et al., 2021). Regenerative business models take a more holistic view of their business practices and aim to regenerate the natural and societal spaces in which they operate (Hahn and Tampe, 2021). They employ their product and/or service offerings to actively regenerate the natural and societal spaces they operate in (Konietzko et al., 2023). Their product or service has additional value added to them due to this characteristic of regenerative operations and supply chains. The business models of such organisations aim to regenerate the natural and societal systems they operate in, creating value for all stakeholders, instead of just shareholders (Konietzko et al., 2023). An example of this is Guayaki which sells Mate Yerba produced using indigenous practices, and in collaboration with local communities, such that the production process revitalises local ecosystems (Guayaki, 2023). Another example is Playa Viva in Mexico that incorporates the concept of an ecotourism hotel with that of local reforestation using permaculture principles, creation of turtle sanctuaries, and revitalisation of the local Juluchuca community through education and community projects (Playa Viva, 2023). These examples provide a good starting point for defining and understanding the nature of regenerative business models

2.2. Regenerative business strategies

What specific strategies can a business employ to be regenerative? A first conceptualisation of regenerative business strategies that could be operationalised by business was developed by Hahn and Tampe (2021). Moving beyond theoretical discussions and towards operationalising regenerative business models, they described the possibility of regenerative business strategies operating on three different degrees - 'Restore, 'Preserve' and 'Enhance' – in relation to the natural environment. A recent study by Konietzko et al. (2023) conducted an extensive literature and practice review on the subject of regenerative business models, supplemented with insights from focus groups with business practitioners and indigenous communities. We use their definition of regenerative business models, which notes that such business models:

> "*focus on planetary health and societal wellbeing. They create and deliver value at multiple stakeholder levels—including nature, societies, customers, suppliers and partners, shareholders and investors, and employees—through activities promoting regenerative leadership, co-creative partnerships with nature, and justice and fairness. Capturing value through multi-capital accounting, they aim for a net positive impact across all stakeholder levels. (They) recognize that nature is an irreplaceable foundation of human health and wellbeing, that human societies are deeply embedded in the biosphere, and that they depend on the health of the biosphere for their own health*" (Konietzko et al., 2023, p. 375)

Konietzko et al. (2023) also define various principles for a company to become regenerative. Firstly, this involves moving from shareholder to stakeholder profits, generating value for all stakeholders rather than just shareholders (see also Stubbs and Cocklin, 2008). Secondly, regenerative businesses allow for value capture across natural, social, and cultural capital, at every stage of production (see also Quarshie et al., 2021; Vlasov, 2021). They do this by taking a broader perspective on health and recognize that the health of people and organisations is closely linked to the health of nature (Gardner et al., 2022). They could then incorporate this into their business operations by for example designing product and services that improve the health of the planet. Rather than just measuring the negative impacts of business, (such as the environmental footprint), regenerative business models also look at the positive changes their products and services can create (their handprint). The handprint being a measure of substitution that is the difference between the footprint of an offered product or service, and the footprint of a product, service or behaviour that the former substitutes or replaces (Norris, 2015; Norris et al., 2021). Thirdly, regenerative businesses move from false to true pricing of products by internalising all externalities that go into producing products and services (see also Michalke et al., 2022; Roland and Landua, 2015). Regenerative business models take a more holistic view on rights and seek ways of doing business that acknowledges and respects human and nature rights (Polman and Winston, 2021; Sanford, 2017). In doing so, the businesses can become more risk resilient in a world where supply chains are increasingly affected by climate disasters.

More recently, Gualandris et al. (2024) have built on these conceptualisations to propose three new principles for regenerative supply chains: proportionality, reciprocity, and poly-rhythmicity. Proportionality refers to the adjusting scale and scope of production and consumption within the boundaries of socio-ecological systems. Reciprocity is about interactions between supply chain stakeholders, local communities, and nature being designed to benefit both human and non-human actors. And finally, poly-rhythmicity refers to working with the biorhythms of the natural spaces that supply chains operate in.

What then is the difference between a regenerative business model and one that is characterised as a sustainable or circular business model? According to the review by Konietzko et al. (2023), the key difference lies in the core perspectives that the three take while approaching business and the environment. Sustainable business models focus on reducing the footprint and resulting harm of a business model. Circular business models strive to close, slow and narrow resource loops, making business and consumption systems as resource efficient and self-sustaining as possible (Bocken et al., 2016). Regenerative business models take a more holistic approach, by focusing on the role of economic activity within ecological systems and how that activity can contribute to the health and prosperity of the ecological system (Hahn and Tampe, 2021; Konietzko et al., 2023). Thus, while regenerative business models are essentially a subset of sustainable and circular business models (Konietzko et al., 2023), it is important to delineate them from the others as they are distinct in their functional approach and level of ambition. However, similar to sustainable and circular strategies, it is important to have transparent impact reporting of regenerative business strategies. Without transparent impact reporting, businesses that are not genuinely regenerative and are simply greenwashing, could end up outcompeting truly regenerative businesses.

2.3. Greenwashing and materiality in relation to regeneration

While a promising idea, the 'regeneration' label can also be susceptible to intentional or unintentional greenwashing (Gordon et al., 2023). Greenwashing refers to deceptive or misleading marketing practices that exaggerate or misrepresent the sustainable goals and actions achieved by a company (Griese et al., 2017). This can be done to improve profits and/or reputation by presenting an environmentally responsible image (Griese et al., 2017). This can include making a greater effort to promote the sustainable agenda or record of a company than is spent on actual sustainable actions (Nemes et al., 2022). For example, in 2012 Mondelez International, the owner of popular chocolate brand Cadbury, launched the Cocoa Life Initiative that aimed to restore deforested land, create better supply chains around cocoa farming, and eradicate child labour (Mondelez International, 2024). The project was widely marketed as an initiative that would provide fairer compensation to cocoa farmers and restore deforested land in the farming regions. However, closer scrutiny by journalists in 2022 revealed lax monitoring practices and rampant exploitation still existing (Channel 4 UK, 2022; Ungoed-Thomas, 2022). Another example is Coca Cola which once claimed 'water neutrality' whereas it barely accounted for its operational water use and excluded the impact in its wider supply chains which had left farmers e.g. in India with empty watersheds (MacDonald, 2018).

However, we do not believe that the risk of greenwashing for the term regeneration is greater than those of any other terms associated with sustainability in business such as, circularity, recycling, or indeed 'sustainability' itself. In all cases there is a need for stronger policy and regulations to ensure that misuse of these labels is avoided. Such policy measure proposals are already underway, for e.g. in the EU and the UK to create stricter anti-greenwashing laws (European Commission, 2023; FCA, 2023; Government of UK, 2023). Recently the UK advertising authority announced that it would clampdown on misleading 'biodegradable' and 'recyclable' claims (Greenfield, 2023). The effect of such policies in inhibiting greenwashing though still remains to be seen. Further, studies have also shown that consumers and employees are not always naïve to greenwashing efforts, and engaging in greenwashing can damage a firm's reputational standing with its customers (Ioannou et al., 2022; Robertson et al., 2023). In fact, Robertson et al. (2023) found higher perceived greenwashing to be positively correlated with perceptions of corporate hypocrisy, and in turn the likelihood of employees' intentions to leave an organisation (which increases operational costs for business in the long-term). Consequently, there is a need to use strict inclusion criteria when investigating cases in the regenerative business model sphere.

Another risk with creating regenerative business models is that of rebound effects that undermine the intended environmental gains from the new business models (Zink and Geyer, 2017). These could for example occur because of increased overall consumption due to moral licensing that consumers might experience from purchasing products/ services they perceive as being regenerative (Dütschke et al., 2018). However, accounting for rebound effects is outside the scope of this study, as it would require an independent environmental and societal assessment of each individual case, and thus it is not accounted for.

2.4. Research gap

While promising, regeneration may also appear to be a challenging business strategy. First, it may require customers that are willing to perhaps pay more than they would for a non-regenerative product or service (Bastounis et al., 2021; Herrmann et al., 2022). Second, regeneration of nature and societies requires skills and knowledge that is typically beyond what is considered conventional business skills. It may entail collaborations and partnerships with NGOs, local communities and governments or nature conservation agencies (Hardman, 2009; Polman and Winston, 2021). Collaborations like these, that are aimed at regeneration rather than profit, are predicted to be increasingly important, since business would take on a greater societal responsibility than they have traditionally, to benefit nature and society (Hardman, 2013; Polman and Winston, 2021). Thus, there is an apparent need to make regenerative business models more accessible. Past literature has also highlighted several gaps in research of regenerative business models, such as how regenerative business models might be enacted, how they relate to their broader socioeconomic context, and whether certain types of regenerative strategies are more likely to be employed by smaller or larger companies (Hahn and Tampe, 2021; Konietzko et al., 2023). This research seeks to answer some of these questions by creating a comprehensive database and typology regenerative business strategies. By collecting examples of viable regenerative business cases to illustrate how businesses can achieve regeneration, regenerative business models may become more accessible to entrepreneurs and business designers, while also addressing some of the questions raised by the literature.

3. Method

3.1. Data collection: regenerative business case database

The development of the regenerative business case database and typology first involved an iterative practice review process to collect regenerative business case examples. Practice reviews can be a useful tool to gain insights into the current state of business practice, how alternative strategies are implemented, and the outcomes of their effectiveness. This can be especially useful when business practice appears to be ahead of academia (Bocken et al., 2014). As no directory of regenerative business model cases existed to date, the cases were collected through purposive sampling which was conducted in two parts (Bryman and Bell, 2011; Saunders et al., 2009). The first part involved structured searches for regenerative business cases through SCOPUS, Ecosia and Google search engines, and scanning of popular sustainable business case databases. First, a SCOPUS title-abstract-keyword search was done using the following search query: ("regenerat*") AND ("business model*" OR "company case*" OR "business case*"). The resulting journal articles were scanned for relevant regeneration related business cases. Then Ecosia and Google search engines were used to identify regenerative business cases. Keywords such as "regeneration", "restorative", "net positive" were used in conjunction with synonyms for business ("business", "company", "firm", "enterprise", "business models") in multiple combinations. In Google, the first 10 search results were scanned for cases, and in the Ecosia search engine the first search result page was scanned to find business case examples. This inclusion limit was adopted as both search engines generate tens of thousands of search results for any single query, however the relevance of the search results tends to decrease beyond the first page. This was then followed by scans of popular databases of new business models such as the Ellen MacArthur Foundation, Greenbiz and The Blue Economy, databases also identified in the study on regenerative business models by Konietzko et al. (2023). All searches only included cases described in the English language. The complete search protocol, keywords used for search queries, and resulting outcomes can be found in Appendix S1.

The second part of data collection involved supplementation of cases through expert recommendations from other researchers, as well as cases identified through newsletters and networks that the authors were part of. The initial search was conducted between June and September 2023, and the supplemental cases were added continually during the writing of this paper until February 2024 in order to have as inclusive a database as possible. Fig. 2 in Section 3.2 shows the detailed data collection and analysis process.

The primary inclusion criterion was that businesses must describe themselves or some aspect of their business practices as regenerative (or synonyms such as restorative, net positive, etc.) in their website and/or online communications. The second inclusion criterion was that the business cases needed to show active ongoing regeneration efforts, in order to exclude cases that simply had hypothetical goals. The third criterion was that the cases needed to have active revenue generation activities, in order to exclude cases that were not active in the market yet and so did not have a proven business model. The final and most important filtering criterion was that the companies needed to show transparent impact measurement methods regarding their regenerative activities. Companies that did not transparently report on their impact measurement were excluded from the database. Insofar as possible, it was ensured that the included companies had independent third-party verification of their claims (E.g., certifications, audits, etc.). When third party verification was unavailable, concrete, detailed reporting (for example with before-after images, videos and statistics) was deemed acceptable as an inclusion criterion. This was done to ensure that smaller companies that sometimes do not have the resources to pay for expensive certifications and audits (Das et al., 2022) are not unduly excluded. These criteria were developed based on expert feedback and consensus in the author team, with the main aim of ensuring robustness of case examples. Table 1 further details the main inclusion criteria used.

The first part of the data collection resulted in a total of 118 business case examples, which were then scanned for duplicates (44), and relevance based on the criteria identified in Table 1 (resulting in 25 deletions). This was then supplemented with 36 additional cases through expert recommendations during the writing of this paper. This resulted in a final regenerative business case database of 84 cases, the full list of cases can be seen in Appendix S2. On these resulting cases secondary data was collected through grey literature available on company

Table 1
Main criteria for inclusion of business cases based on literature.

	Description	Examples	Sources
Communication of regenerative activities	Communication of aspects of the business model that are regenerative (or similar synonym)	Greenwave states as its purpose statement: "*Greenwave replicates and scales regenerative ocean farms to create jobs and protect the planet.*"	(Bocken et al., 2022)
Continued / ongoing regenerative effort	Ongoing projects with proof of concept instead of conceptual idea or goal.	Mossy.earth works to restore degraded ecosystems in evidence-based ways. They obtain their ongoing funding through members subscriptions and donors and produce periodic in-depth progress reports that are publicly accessible.	(Bocken et al., 2014)
Has active revenue generation activities	Ongoing revenue generation activities, in order to exclude organisations who would not be considered to have a proven business model	The Ocean Cleanup which cleans up plastic in the ocean, is a registered as a non-profit organisation. But they generate revenue through leasing of their plastic collection technology to municipalities and through sale of products made from the plastic they harvest from the ocean.	(DaSilva and Trkman, 2014)
Shows proof of positive impact	Measured impact claims that are transparently communicated.	Chuk produces compostable tableware made from sugarcane waste. They list their various third-party certifications transparently on their website.	(Griese et al., 2017; Montgomery et al., 2023)

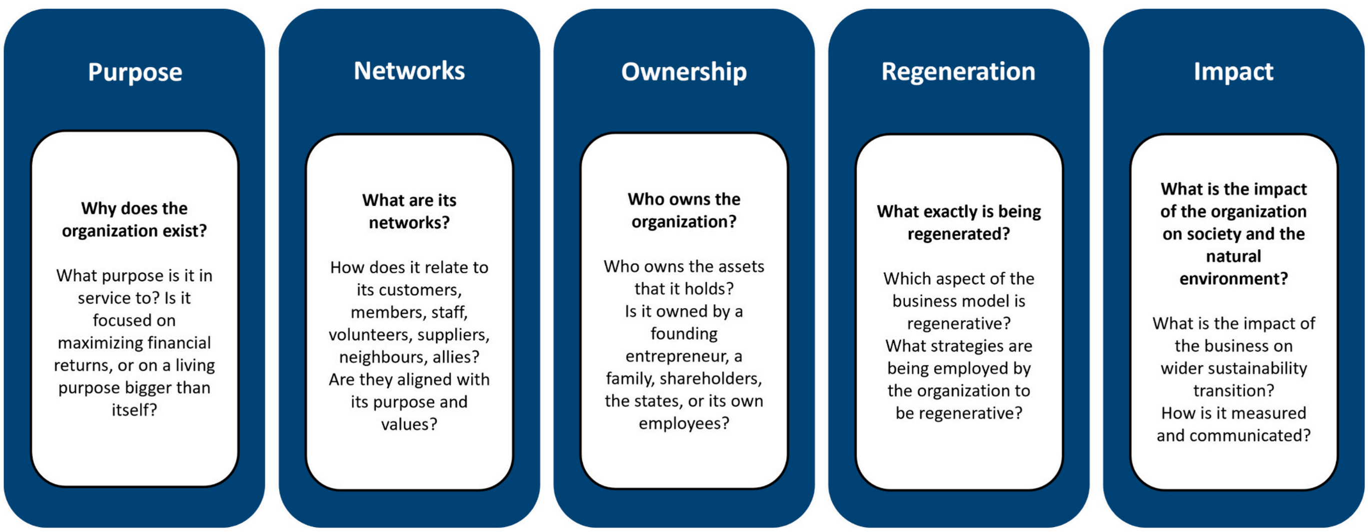

Fig. 1. Framework for identifying core characteristics of regenerative business models (based on framework for business for sufficiency-based economy by Bocken et al. (2022)).

webpages, publicly available reports and newspaper articles. The information collected for the cases was based on the framework of core elements of business models in a sufficiency-based economy developed by Bocken et al. (2022), which was adapted to fit the regenerative business context (Fig. 1). The framework describes six core elements of a business model for sufficiency: Purpose, Networks, Governance, Ownership, Finance, Impact and Scale-up. We decided to merge the 'governance' and 'ownership' elements into one 'ownership' cluster, a replace the 'scale-up' element with a 'regeneration' element instead. This was done in order to have a framework deemed most suited to identify core characteristics of regenerative business models. This involved collection of data that described the companies' business model operations such as purpose, methods of value proposition, creation and capture, types of regenerative strategies employed, ownership structure, and impact claim verification.

3.2. Data analysis: coding to develop a database and typology

The qualitative core characteristics collected about the business cases were then further coded in Excel and Atlas.ti based on the grounded theory method described by (Corbin and Strauss, 1990). This involved application of inductive codes to identify common themes that could be developed into a typology of regenerative business strategies. First descriptive codes on the core operations of the business cases (e.g. sector, founding year, headquarters, etc.) were listed separately in Excel. Second, to create a typology of regenerative business strategies, open coding was done to identify categories such as dominant regenerative innovation themes, regeneration initiatives, and operational processes. The main question guiding this part of the coding process was: *How is regeneration occurring, and what are the regeneration strategies being employed?* This was followed by axial coding in Atlas.ti to identify

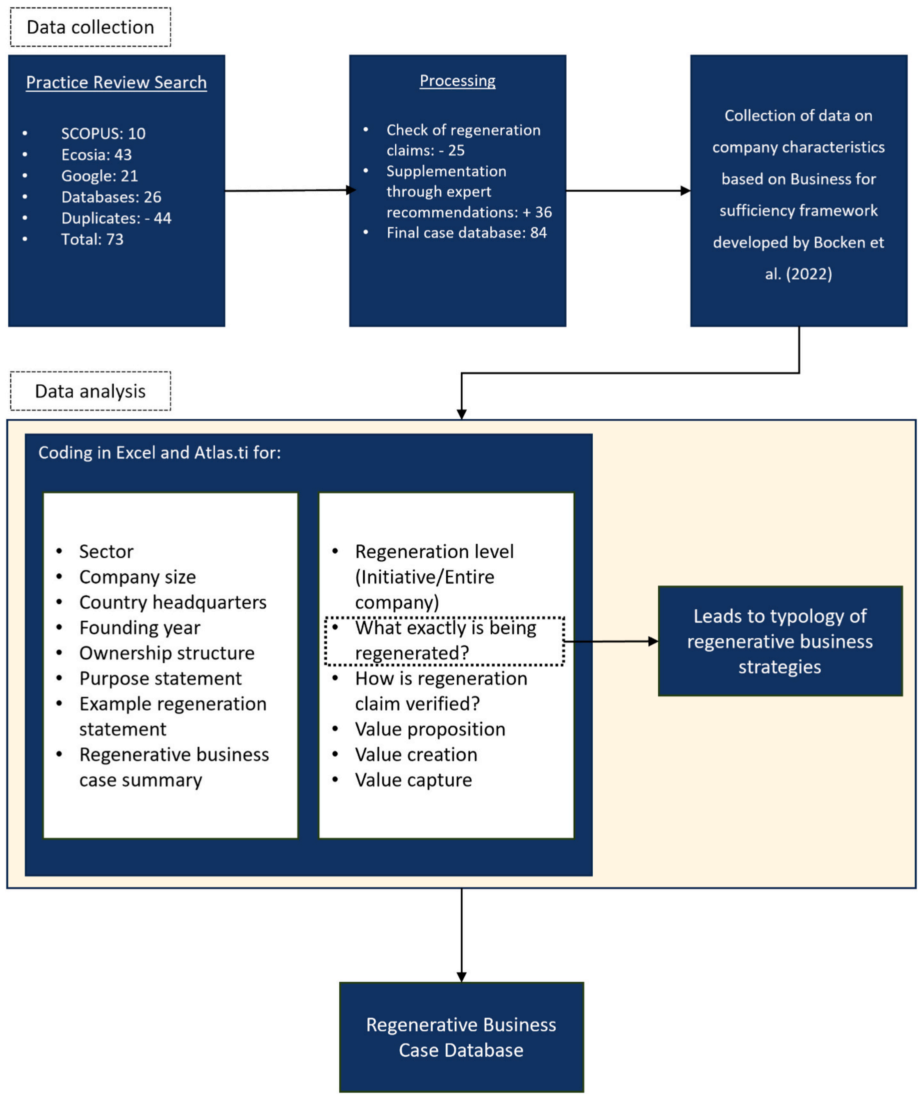

Fig. 2. Methodology followed in data collection and curation process.

interrelationships between the codes, such as similarities or differences between the different regenerative business strategies. Finally selective coding was done, whereby clusters of regenerative business strategies were identified and organised into categories, that resulted in a typology of regenerative business strategies. Both these smaller codes of regenerative business strategy types and overarching archetypes are described in Table 3 in the Results. All the information collected in both parts of the coding process was then input into creating the final regenerative business case database. Fig. 2 visualises the methodology followed during data collection, coding, analysis and creation of the database and typology. The Regenerative Business Case Database can be found in Appendix S2.

4. Results & discussion

The following subsections describe the two primary results of this study.

4.1. The regenerative business case database

The largest share of regenerative business case examples in the database consisted of small companies (34), followed by very large companies (25), large (13), and medium sized companies (12) (Fig. 3). The examples were from 15 sectors, with the largest number of cases coming from the food (22), consumer goods (19) and fashion sectors (18). The business activities of some companies fell into multiple sectors, resulting in a higher final frequency tally of sectors (107) than the total number of cases (84). Fig. 4 shows a breakdown of the various sectors the regenerative business cases operate in. It was observed that in 65 of the cases regeneration was occurring at the level of the entire company and was instilled into the core purpose, missions and/or ethos of the business. Examples of this are Fairafric's aim of only creating chocolate that redirects profits to producers in Ghana (Fairafric, 2024), or True Pigments who only create paint from polluted mine waste that is contaminating rivers (True Pigments, 2024). Whereas, in 19 company cases, regeneration was observed on the level of 'initiatives', that is, in the form of smaller initiatives or strategies within larger business models that might still be in transition. For example, IKEA's regenerative procurement strategy for certain product lines (IKEA, 2024), or Bupa's investments into creating healthier urban living environments (Bupa, 2024), both smaller regeneration initiatives or strategies within their large business models.

The database also revealed that the first regeneratibe business case (Familia Torres) was founded in the year 1837, with progressively more companies being founded since the late 1990's. Perhaps unsurprisingly, there has been an increasing interest in pursuing regenerative business

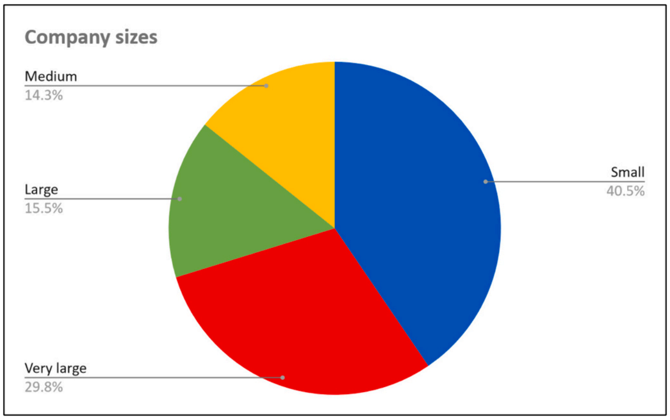

Fig. 3. Size of case study companies based on number of employees (Small: <50, Medium: 50–249, Large: 250–500, Very large: >500).

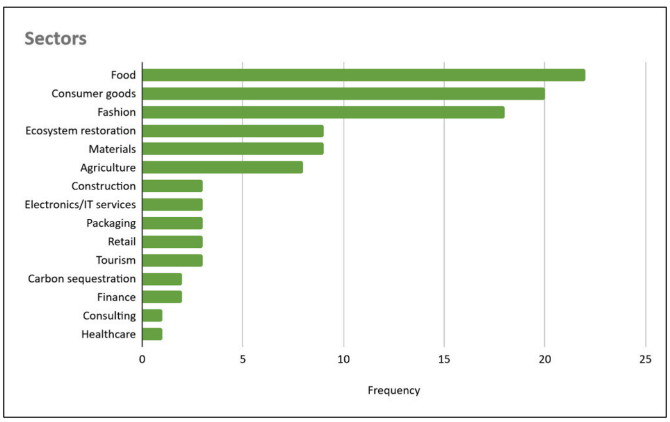

Fig. 4. Sector distribution of case study companies.

strategies in recent years (Fig. 5), due to sustainability being put on the agenda more by governments and companies alike. It was also observed that regenerative companies can be found operating across the world in multiple continents and countries (Table 2). The highest number of cases were found in the USA (39), followed by the UK (12) and the Netherlands (7).

Further, the different ownership patterns of the case study sample can be seen in Fig. 6. The most frequent form of ownership observed was private ownership (31), which often meant that the company was

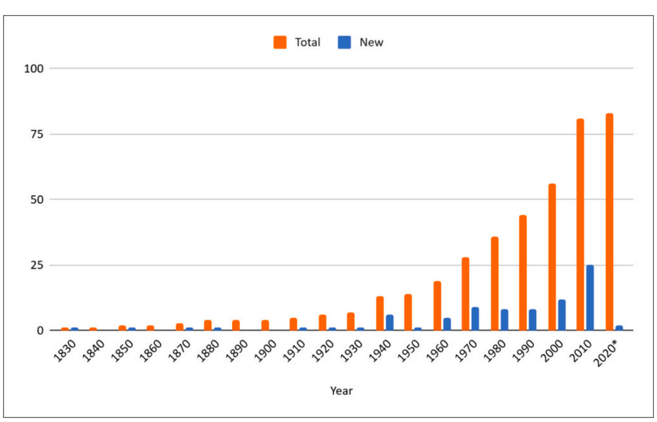

Fig. 5. Number of companies founded per decade (*2020 being an outlier with only 3 years counted, including the Covid-19 pandemic years).

Table 2
Headquarters of the regenerative business cases.

Country Headquarters	Number of companies	
Americas (North & South)	45	
USA		*39*
Brazil		*4*
Canada		*1*
Mexico		*1*
Europe	32	
UK		*12*
Netherlands		*7*
Germany		*5*
France		*5*
Denmark		*1*
Spain		*1*
Switzerland		*1*
Asia	5	
India		*3*
Indonesia		*1*
Japan		*1*
Africa	2	
Ghana		*1*
Kenya		*1*
Oceania	2	
New Zealand		*2*
Total		**84**

usually owned by its founder(s). This could be influenced by the large proportion of small companies within the sample (40.5 %). The second most common case was ownership by shareholders in the stock market (29), half of which were through a holding company. The third most common form of ownership was cooperative ownership (10). The co-operatives category includes one consumer cooperative, two co-operatives consisting of other businesses and five employee-owned cooperatives. In the case of two companies – LUSH and Eileen Fisher – the ownership structure was a combination of two different types (private and employee-owned), this has been counted separately in both categories. Beyond this, there were nine family-owned companies, and five owned by foundations, trusts or associations. We also observed a smaller number of steward-owned companies (2). These are a novel form of ownership structure where dividends are never paid out, and run by stewards in pursuit of a specific purpose or for the benefit of stake-holders, rather than shareholders (Sanders, 2022). While a steward may hold shares or otherwise control and manage the company, they are prevented from profiting from their position and thus do not have motive to maximize profit (Sanders, 2022). This helps ensure that the purpose of the company remains central to its business. Although they make up a smaller portion of the companies within the sample, co-operatives, associations, trusts, foundations and steward-owned companies – ownership structures with similar value footings – taken together were not uncommon (30 %). This could reflect a characteristic of companies that strive for regeneration to be driven by purpose.

Finally, it was also found that companies reported and verified on their regeneration claims in various ways, often using multiple different measurement, verification and/or certification methods. This diversity in methods might be due to the business cases ranging from 15 sectors, with businesses of varying sizes and thus different opportunities for verifying their claims. Table 3 showcases the methods that had a frequency higher than five in the database. A more detailed list of measurement methods can be found in the Regenerative Business Case Database in Table S2–1 in the Appendix. The most common way of reporting impact was through annual sustainability or impact reports. These were often a compilation of various metrics and key performance indicators (KPIs) based on various standards such as the GHG protocol, the Science-Based Targets Initiative (SBTi), the Sustainable Development Goals (SDG), etc. The second most common form of reporting was the B-corp certification. This consists of several standards determined by the B Corp organisation, relating to environmental and social performance transparency and accountability in order to "positively impact all

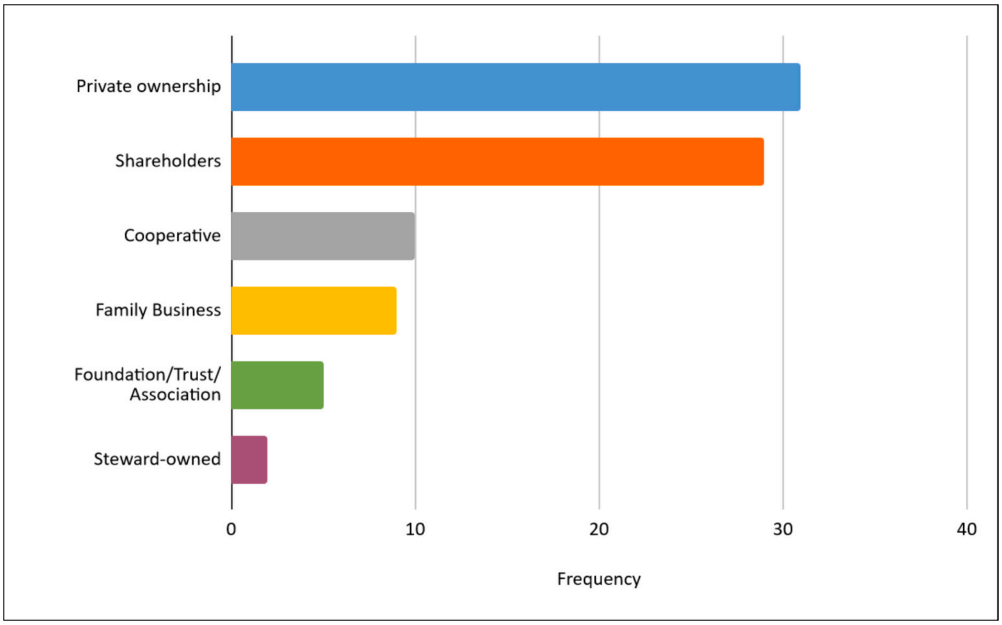

Fig. 6. The prevalence of different ownership models within the sample database.

Table 3
Various ways in which regeneration claims are reported on by the companies in the database.

Impact reporting method	Frequency
Own sustainability/impact report	65
B-corp certification	30
Various organic certifications (such as Regenerative Organic Certification (ROC), USDA organic, Ecocert Organic, Organico Brazil, Regenerative Organic Cotton, NOP Organic Cotton, etc.)	22
Own KPIs (decided on with stakeholders)	18
KPIs based on GHG protocol (scope 1–3)	17
Measurement through third-party research institute/NGO/association	17
Video, Image, GPS, Map, Project Updates	15
GRI standards	13
Third-party audits	13
Fairtrade certifications (such as Fair Trade certification, Fair Trade Federation, Fair for life, Natureland Fair, Fair Trade Proof, Fair Trade International)	12
KPIs based on SDGs	12
ESG Reporting	9
KPIs based on Science-Based Targets (SBTi)	8
Regenerative Organic Certification (ROC)	8
ISO standard	7
Life-cycle assessment (LCA)	7
SASB standard	7
Task Force on Climate Related Financial Disclosures (TCFD) Reporting	7
Responsible Wool/ZQRX Certifications	6
Materiality assessment	5

stakeholders – workers, communities, customers, and our planet” (B Corp, 2024). The third most common form of impact reporting was through various organic certifications such the Regenerative Organic Certification (ROC), USDA organic, etc. In total there were 60 different types of reporting or measurement methods used by the companies. This shows that there is no one standardised way of measuring impact of regenerative business strategies yet. However, this is in line with the diversity of strategies used, with relevant metrics employed rather than all measurements being simplified into one single metric, such as CO_2 emissions. This is in line with past research findings on how companies measure environmental impact of new business strategies (Das et al., 2022).

4.2. Typology of regenerative business strategies

The second contribution of this study is a typology of regenerative business strategies (Table 4). It was found that the different types of regeneration strategies followed by businesses could be broadly classified into six clusters which are: regenerative leadership, nature regeneration, social regeneration, responsible sourcing, human health & well-being focused, and employee-level focus. These are further described below and in Table 4.

Regenerative leadership: The regenerative leadership cluster comprises strategies that lay the foundation for regeneration in a broader sense throughout the company. One example is making the supply chain transparent, as in the case of Community Grains, who detail individual harvests and the farmer responsible for the harvest. By making the origin of their raw materials open to scrutiny, companies can help consumers make better informed decisions about the goods they are purchasing, as well as allowing them to ensure that the product does in fact contain ‘what it says on the tin’ (Community Grains, 2024). Another common strategy is to support regenerative practices within an already existing supply chain. For example, ethical cosmetics brand LUSH actively use their purchasing power and supply chains to achieve regenerative outcomes, such as the protection of biodiversity, or rewilding projects (LUSH, 2024a). Helping suppliers transition to regenerative practices can be a good way of changing the impact of an existing value chain and business model. Meanwhile, the reinvestment of profits into nature and society, and advocacy are ways in which businesses can use their profits and social influence to regenerate the world around them.

Nature regeneration: Nature regeneration consists of a series of strategies that aim to improve the health of the natural world. For example, regenerative agricultural practices, such as those employed by Native in their sugar cane plantations, can be deployed to increase the health of the soil, improve biodiversity and build stronger and healthier ecosystems (Native, 2024). Regenerative practices can also be deployed in marine ecosystems, such as the case of CarbonEthics, who restore mangroves in Indonesia which in turn restores marine and coastal ecosystems while also sequestering carbon (CarbonEthics, 2024). Another example is through restoring or conserving forests, or other natural spaces so that flora and fauna can thrive. Consumer electronics manufacturer Ricoh have forest conservation initiatives in Japan and plans to

Table 4
Categorisation of regenerative business strategy archetypes.

Regenerative Business Strategy Archetypes	Type of strategies supporting regeneration	Frequency in data sample (x out of 84)
Regenerative leadership	Transparent verification of claims	22
	Creating new regeneration pathways in supply chain	18
	Reinvesting profits for planet & people	6
	Advocating for equity & justice	5
Nature Regeneration	**Biodiversity regeneration**	**42**
	Soil health/land regeneration	**39**
	Ecosystem restoration	**32**
	Water body restoration/sequestration	18
	Carbon sequestration	17
	Forest restoration	13
	Marine regeneration	11
	Forest conservation	6
	Biodiversity conservation	6
	Animal welfare	5
Social regeneration	**Community regeneration**	**36**
	Creation of alternative livelihoods – climate adaptation	18
	Involving local communities in decision-making	13
	Just labour conditions	3
	Connecting stakeholders to nature & communities	4
Responsible sourcing	**Traceable supply chain**	**23**
	Fairer compensation for farmers	20
	Supporting small businesses	11
	Long term contract commitments	6
	True pricing	2
Human-health & well-being focus	Human-health & well-being regeneration	9
	100% nature-based products	6
	Urban green spaces restoration	2
Employee level focus	Fair wages for workers	9
	Just labour conditions	3
	Increasing employee involvement in decision-making	2
	Improving employee health & wellness	2

plant 1 million trees by 2030 (Ricoh, 2024). Finally, protecting, and reintroducing endangered animal species can contribute to ecosystems in important ways that allow biodiversity to increase and the ecosystem to grow stronger over time.

Social regeneration: While nature regeneration is aimed at regenerating natural ecosystems, social regeneration is aimed at regenerating human ecosystems. Prominent examples include fair-trade practices, such as those employed by organic soap maker, Dr. Bronner's sister company, Serendipalm who partner with smallholder farmers in eastern Ghana to regeneratively produce palm oil, while providing fair remuneration and education (Serendipalm, 2024). Another example is Fairafric who place manufacturing facilities of chocolate in Ghana to generate more wealth where cocoa is grown (Fairafric, 2024). Their overall goal being to increase the concentration of wealth within the producer and farmer part of the supply chain and moving it away from the distribution and manufacturing parts, where wealth otherwise often resides. By giving farmers and farming cooperatives a bigger piece of the pie, otherwise impoverished communities get a chance to grow and thrive. Since farmers often inhabit former colonies and the distribution side of the supply chain tends to reside in former colonial nations, this reverse flow of wealth can be an important way of reversing historical power dynamics. By involving these same communities in the decision-making process, providing just and equitable working conditions businesses can create the conditions for communities to thrive. Thus, through the generation of wealth and prosperity, a mutually beneficial symbiotic relationship can flourish between communities and their natural resources. This also creates the preconditions for a sustainable economy to thrive, since the basic need for safety and security have been met.

Responsible sourcing: Responsible sourcing entails a series of sourcing practices that help to empower and strengthen suppliers and their communities. By providing fair compensation to farmers, ensuring the traceability of ingredients and raw materials, and supporting small businesses and farmers within a supply chain, businesses can create value and prosperity throughout the supply chain. It can also serve to shrink the distance between consumer and producer, such as in the case of Albert Heijn's True Pricing scheme, where a consumer can choose to pay the "true" price of a product, as it would be if everyone was fairly remunerated all along the supply chain (Albert Heijn, 2024). Long term commitments between businesses and their suppliers, such as those between the Cooperative Coffees roasteries and the cooperative farming communities that supply them, can also help reduce risks that might otherwise hinder economic investment within a supply chain (Cooperative coffees, 2024).

Human health & well-being: A focus on human health and well-being entails creating product and services that have a positive effect on their customers. Natural, organic and, or safe, ingredients that make people healthier and happier or indeed urban spaces designed to make human lives better and healthier are examples of a focus on human health and well-being. Examples include Claytec's natural clay building materials that create healthy living environments (ClayTec, 2024), or health insurance provider Bupa's Healthy Cities initiative that aims to promote better human health so that there is a smaller illness load on healthcare services (Bupa, 2024).

Employee level focus: Businesses that have an employee level focus employ strategies to improve the working conditions, lives, and well-being of their employees. While this perhaps most prominently includes fair (or living) wages and more just labour conditions, it can also include a genuine attempt to change the lives of employees. For example, Greyston Bakery's hiring and reskilling of former convicts and addicts to help them on to new careers (Greyston Bakery, 2024). It can also mean making employees part of the decision-making process, as is seen in the case of cooperative businesses like Dean's Beans, or LUSH's employee ownership schemes (Dean's Beans, 2024; LUSH, 2024b). Finally, providing services that contribute to employees physical and mental health is another way of focusing regenerative efforts on the people who make up a business.

5. Discussion

Below we discuss the types of regenerative business strategies that have emerged, the extent to which a business can be regenerative, the contributions of this study to research and practice, and finally the limitations of this study.

5.1. What regenerative business strategy types have emerged?

In this section we discuss the types of organisations, the dominant sectors, the partnerships of regenerative businesses, and pathways undertaken to pursue regeneration. Table 5 provides a brief overview of the dominant organisational characteristics of the dataset.

First, regarding the relative sizes of the organisations within the regenerative business case database, the dominance of both small and very large businesses is clear. This is interesting as it might tell us something about the nature of business model development. On the one hand larger organisations with well-established revenue streams can afford to experiment with new business model strategies that smaller organisations would be unable to support. They may also want to improve their reputations as leaders within sustainable business practices. On the other, small companies and startups have a need to

Table 5
Overview of the main organisational characteristics identified in the 84 regenerative business cases.

Dominant sectors	Food, consumer goods, fashion
Size of companies	Small: 40.3 %, Medium: 14.3 %, Large: 15.5 %, Very large: 29.8 %
Main headquarter locations	USA: 39, UK: 12, Netherlands: 7, Germany: 5, France 5
Types of organisational ownership	Private ownership (often founders), shareholders, cooperative, family business, foundation, trust or association, and steward-owned
Most prevalent types of regeneration strategies	Biodiversity regeneration, Soil health/land regeneration, Community regeneration, Ecosystem restoration, Traceable supply chain

establish themselves in the marketplace and can thus benefit from differentiating themselves from the competition and finding novel ways to create value. Working together, they can both accelerate the transformation towards a more regenerative economy (Hockerts and Wüstenhagen, 2010).

Second, in terms of sectors, business models based upon regeneration strategies seem to proliferate in the fairly polluting sectors of food, consumer goods and fashion (CDP, 2019; Crippa et al., 2021; Leal Filho et al., 2022). This also appears to be tied to land-use change, as land is often where value is captured (Lozano-García et al., 2020; Zermeño-Hernández et al., 2016), be it through coffee beans or lumber. This makes sense for several reasons. If there was not a clearly identifiable need for regeneration, businesses by their nature would be unlikely to attempt it. Further, sectors with a lot of pollution would also be the sectors with the greatest potential for development, with more 'low hanging fruit'.

Third, a majority of the companies within the database were also founded in or after the 1990's. This may indicate a trend, with the Brundtland report published in 1987 being the first major public recognition of the effect of human activity upon the earth's ecosystems (Brundtland, 1987), later followed by increasing attention on climate change and biodiversity loss.

Fourth, a clear trend within the database is that many organisations had partners to implement, be it public sector, NGO, or other businesses. This may imply that regeneration is not something that an organisation can simply do on its own. Much of the required expertise required for regeneration might not reside within the business, and issues may transcend well beyond company boundaries. Rather, regeneration requires engaging multiple parties and stakeholders in collaborative effort, for it to be successful (as also observed in Konietzko et al., 2023). This ties in with another observation, that usually multiple regenerative strategies were employed simultaneously (typically two or more), and they tended to tie into each other, improving their mutual effects through positive spillovers (Elf et al., 2019). Given the holistic nature of regeneration, this is to be expected. In line with this, although not considered within our typology, the regenerative business strategy types that have emerged can be considered to effect regeneration directly or indirectly. Some strategies, such as practicing regenerative farming or conserving biodiversity could be classified as a direct regeneration enabling strategy. On the other hand, strategies such as increasing traceability within a supply chain or transparent reporting and verification of regeneration claims can be considered more indirect, regeneration supporting strategies. It could be argued that the main function of the more indirect strategies is to support the more direct strategies. Taking the 'transparent verification of claims' strategy as an example, it was found by Konietzko et al. (2023) that transparency performs the important function of building trust and ensuring accountability in regenerative businesses. Therefore, it can be considered an essential component of regeneration. A similar argument can be made with regard to involving local communities in decision making. It allows stakeholders to be part of determining what supports their own development and thus serves the purpose of regeneration. While these different types of direct-indirect strategies might fall on a spectrum, it is important to consider them as part of a whole, working together and supporting each other.

Fifth, companies that showed regenerative leadership were often actively engaged in creating new regenerative supply and, or value chains that did not exist before. They showed initiative in exploring pathways that others had not explored before. Gualandris et al. (2024) have pointed to the need for companies to use their supply chains to further the cause of sustainability. The supply chain initiatives observed within the cases in the database are very much in line with that ambition. Large and very large companies that were in the process of transition often achieved their regenerative initiatives through partnerships with sister companies, foundations, and, or cooperatives in more degraded regions of the world. These apparent trends imply that in order to be successful, regenerative business strategies need to be embedded in spaces and communities (Hahn and Tampe, 2021). Humans and their social structuresare often construed as separate from the earth's ecosystems, when in fact they are very much a part of natural ecosystems. Thus, local communities can and need to be leveraged to achieve regeneration within the ecosystems that they inhabit.

Sixth, it was observed that impact verification of regeneration claims was done through a variety of methods, most commonly through annual sustainability or impact reports, B-corp certification or various organic certifications (USDA Organic, Regenerative Organic Certification). However, in all, there were 60 different types of impact verification and reporting methods observed. In other words, there were different ways for companies to measure and verify their impact. This makes sense, as different businesses will have different relevant metrics, requiring different measurement methods; for example, number of people reintroduced to the workforce through job creation can be an important metric, even though it cannot be measured in terms of greenhouse gas emissions. Considering the plurality of contexts in which businesses exist, it may not be meaningful to evaluate the performance of all regenerative business models in the exact same way.

Finally, to conclude, this may mean that, despite apparent trends towards certain regenerative business models (e.g. regenerative agriculture, sourcing, etc.), there is no one size fits all solution or strategy. Every solution needs to be adapted to the ecosystem and context within which it will serve its function. Every regenerative business model being highly dependent on the context within which it operates. Therefore, regenerative businesses may need to be attentive and adaptive, not just to the current needs of the ecosystems they serve, but also to future needs, which may be different. Only through adapting to the current context and ecosystem, can a long-term resilient transition to a regenerative economy be realized.

5.2. To what extent can businesses take regenerative actions?

A common apprehension while approaching experimentation towards regenerative business models might be that companies cannot be regenerative and have a positive impact as they have a focus on profits and exist in the current capitalistic system. Or that focusing on regenerative business strategies will cause companies go out of business eventually as they will not generate enough revenue to remunerate their employees and stakeholders. This study moves towards dispelling this concern, as it was observed that the oldest company in the database that considered regeneration as a core value was the Familia Torres winery, founded in the year 1870, and is still profitable and operational today (Familia Torres, 2024). Their motto has been *"The more we care for the earth, the better our wines"* through five generations of owners. This is important as at times the notion of this cynical myth can act as a deterrent to innovation and moving forward in the transition to a better socio-economic system.

The regenerative business case database also shed light on many good initiatives that can co-exist in countries that are overall not perceived as very sustainable. For example, businesses in the USA are often perceived as lagging behind on climate action when compared to Europe, especially as parts of the country are currently experiencing an ESG backlash (Vanham, 2024). However, we observed that the highest number of regenerative companies (39) are in fact headquartered in the USA (see Table 2).

It should also be noted that there are plenty of examples within the database of companies who do not have regeneration at the heart of their business yet and thus have further space to develop into fully regenerative businesses (for e.g., Unilever, P&G, Danone and IKEA). While it might be tempting to dismiss these companies' contribution, it is important to highlight that it can take time to transition pre-existing linear supply chains and business models to regenerative practices. The steps taken by these companies to trial and develop regenerative business models should therefore be seen as vital first steps towards regeneration, rather than being immediately dismissed as just doing the bare minimum. Often such smaller strategy changes might represent the work of well-meaning individuals or smaller teams working as change agents to build momentum for change within a larger company (Caldera et al., 2022; Visser and Crane, 2010).

Finally, it must also be acknowledged that companies are often the subject of corporate greenwashing (Montgomery et al., 2023), something that regeneration lends itself to too. Indeed, a key takeaway from this analysis was the significant level of greenwashing in using the regeneration label. This was evidenced by the large number of rejections (25) from the initial sample. Therefore, checking for the presence of transparent quantitative impact reporting and verification of claims is crucial before publicising a company as regenerative in articles or blog posts, to prevent amplifying any potential greenwashing (Griese et al., 2017).

5.3. Contributions to research

The regenerative business case database and typology provide multiple contributions to research and practice. From a research perspective, this study adds to previously identified gaps in regenerative business research by addressing questions such as how such business strategies are enacted and how they relate to their socio-economic context (Hahn and Tampe, 2021; Polman and Winston, 2021). Through a bottom-up case study research approach, this study also reconfirms findings in past literature about different types of regeneration strategies that can be employed by businesses, specifically in relation to the restore-enhance-preserve strategies of Hahn and Tampe (2021), and the regenerative principles of Konietzko et al. (2023). However, this study also sets itself apart from these previous studies by creating a database and typology of regenerative business strategies through an empirical practice review of 84 different regenerative business cases, across 15 sectors. In doing so, it gives detailed insights into the organisational purpose, collaborative networks, value proposition, creation and capture models, ownership structures, and impact verification strategies of these various regenerative business cases. This can be used by future researchers to conduct in-depth analyses of cases to identify trends and patterns that allow for the creation of regenerative businesses. The findings also lend themselves towards creation of more sector-specific analysis, and cross-case comparisons of regenerative business cases, and the cases can also be used as a resource in education and training. Finally, the case database also makes a strong case for more research on more radical sustainable business strategies by showcasing that these can also be successful in practice.

In order to further explore how the contributions of this study compare with those of previous studies, we compare the results of this study with those of highly cited past research on regenerative business strategies in Table 6.

5.4. Contributions to practice: guiding business model experimentation towards regeneration

From a practical standpoint, the regenerative business case database offers real-world examples of companies and organisations incorporating more regenerative and socially responsible practices into their business models. This can inspire other companies and allow them to learn about established practices of others, the practical advantages of adopting such strategies for building resilience from a business standpoint, and leapfrog into integrating regenerative practices into their own operations. The practical examples may also alleviate fears of embracing regenerative strategies by showcasing examples of companies regenerating the environment and society, while also remaining financially viable enough to support their business and employees. In doing so, they maintain long-term resilience, and can weather external shocks more easily. The database can also support benchmarking by allowing businesses to assess their current operations and identify areas for potential growth into regenerative strategies. Finally, the database can be used by funding organisations to identify businesses that employ regenerative business practices, that align with their impact goals, and thus increase investment into such companies.

As designers and companies innovate towards regeneration, visions will be needed to set the direction of the innovation process. Changing a business model towards integrating regeneration strategies can begin by asking some key questions about the value creation and capture of the business model to kickstart the process of ideation (inspired by Hawken (2021) and Polman and Winston (2021)):

1. How can we do *more good* rather than just *less bad* as a business?
   a. Do our actions create more life or reduce it?
   b. Do our actions enhance human well-being or diminish it?
   c. Do we create livelihoods or eliminate them?
   d. Are we serving human needs or manufacturing human wants?
2. How can the (natural and social) world be made a better place because our business is in it?
3. Who can we collaborate with to achieve these goals?

Such questions could stimulate initial discussions on how a business may start tackling regeneration as part of its vision and core strategies. While experimenting with regenerative business approaches companies could use the regenerative business case database and typology for

Table 6
Comparison of key findings of this study with past research on regenerative business strategies (*Not as highly cited as the other studies at the time of writing, but we still include them as they are relevant and research on the topic is new).

Study	Geographical scope	Sector focus	Methods used	Outcomes	
				Framework and strategies	Examples from practice
Moreno et al. (2016)	Businesses worldwide	General	Literature review	Proposes a framework for enabling circular business model design that could enable a regenerative system.	Does not give examples of regeneration from practice.
Cetin et al., (2021)	Unspecified	Built environment	Literature review and workshops	Outlines four regenerative strategies in the built environment sector.	Does not give examples of regeneration from practice.
Konietzko et al. (2020)	Businesses in the Netherlands, Germany, USA and Canada	Electronics, energy, ecosystem services, food, packaging, mobility	Literature and practice review, design science research	Proposes a circular business ideation tool, with examples of slow, close, narrow and regenerate business strategies	Provides 10 brief regenerative business strategy examples from practice, and one hypothetical example of a regenerative food ecosystem as part of ideation tool.
Hahn and Tampe (2021)	Businesses worldwide	Food, electronics, tourism, retail, manufacturing, and extractive industries	Theoretical paper developed through a literature review	Proposes principles and criteria for regenerative business, defining three main regenerative business strategies: - restore, such as reforestation after cutting down trees, - preserve such as maintaining the health of an ecosystem while extracting resources from it - enhance by improving the health of an ecosystem.	References 14 regenerative business strategy examples to better explain the 3 types of regeneration strategies.
Konietzko et al. (2023)	Businesses worldwide	Healthcare, mobility, construction, fashion, food, consumer goods, online services	Literature & practice review + focus groups	- Defines regenerative business models by tracing the origins of the regeneration concept - Outlines a framework for principles of regenerative business - Differentiates regenerative business models from circular and sustainable business models	Briefly describes 23 business model examples to better explain the proposed principles of regeneration.
Caldera et al. (2018)*	SMEs in Australia	Electronics, mobility, industrial and agricultural products manufacturing	Qualitative interviews	Outlines a regenerative business strategy for SMEs and the advantages to employing such a strategy and fitting the regenerative strategy into an existing model for the evaluation of sustainable businesses.	Describes practices undertaken by 2 businesses to integrate regeneration into their business practices.
Caldera et al. (2022)*	SMEs in Australia	Industrial and agricultural products manufacturing	Qualitative case studies	Gives an action framework outlining pathways for transition of SMEs towards regenerative business models & identifies key enablers for the transition, specifically having advocates of regeneration within the organisation.	Describes the transition of 2 businesses towards regenerative practices.
This study	Businesses worldwide	15 different sectors: food, consumer goods, fashion, ecosystem restoration, materials, agriculture, construction, electronics/IT services, packaging, retail, tourism, carbon sequestration, finance, consulting and healthcare.	Practice review	Provides a typology of regenerative business strategy archetypes derived from 84 practical examples.	Conducts in-depth analysis of 84 regenerative business cases, which are compiled into a curated database, with data on company purpose, company size, headquarters, business model value proposition, creation and capture strategies, ownership structure and impact reporting.

inspiration. Fig. 7 shows the business model innovation process based on Geissdoerfer et al. (2017) and Bocken and Snihur (2020). Additions in green describe how the database of regenerative business cases and strategies can be used during key stages of the business model innovation process. First, in the ideation phase the database can be used to compare the state of one's business to that of more regenerative businesses. The difference between a firm's own business and regenerative businesses in a similar sector can inform a vision for where to take a business and the goals to accompany it. The database and typology of regeneration strategy archetypes can then be used to help choose strategies that will allow a business to achieve the goals that have been set, while adapting each strategy to the context of the business. In the experimentation phase these strategies can then be implemented in collaboration with all stakeholders, piloted, and continuously evaluated and improved upon. Key performance metrics can be chosen with the help of the regenerative business case database as well as how to measure and record them. As the regenerative business strategy is evaluated, the database can once again be used for bench-marking and comparison.

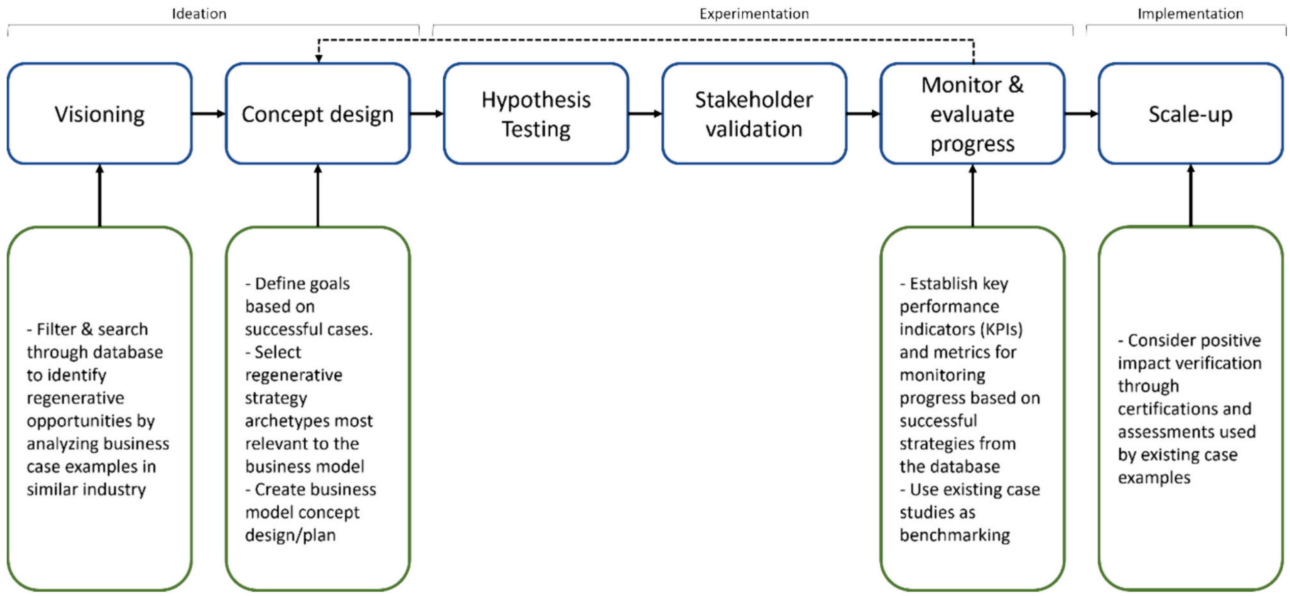

Fig. 7. Using the regenerative business strategies database and typology during business model experimentation (based on Geissdoerfer et al. (2017) and Bocken and Snihur (2020)).

Finally, in the implementation phase the strategy can preferably be scaled up and positive impact verified through assessments and certifications like the ones found in the database.

Thus, the regenerative business case database and strategy typology has the potential to aid business innovators in informing several steps in the development of regenerative business models. Future research could develop business model development tools that integrate the use of the database, which could also update and expand the database to better reflect future practices.

5.5. Limitations

This study had a few limitations. First, the cases included in the Regenerative Business Case Database were identified via purposive sampling. Purposive sampling is a non-probability sampling technique and is susceptible to sampling bias which can skew results due to its reliance on existing networks. This method's inherent lack of randomness and control over sample size could challenge the generalizability of its results (Saunders et al., 2009). However, we attempted to remedy this by collecting cases until data saturation was reached, i.e. until additional cases did not yield any new regeneration strategy codes.

Second, the database utilises data from the companies' official webpages to build the company case studies. This was done in order to include data that is easily accessible. Attempts were made to ensure the quality and reliability of the cases by only including cases whose claims had been verified by empirical studies or through third-party certifications. However, since the information referenced lives on the internet, there can be future issues with inaccuracies resulting from the dynamic nature of online content. Web pages can be taken down or changed, or links can become broken over time, leading to the loss of valuable data, making it difficult for others to verify or replicate the research in the exact same way. This is to be expected as change is a natural part of business and it's reasonable to expect the companies within the database to change as well. For example, towards the end of writing this paper, Unilever announced they were scaling back on sustainability pledges in order to cut costs (Davies, 2024). This limitation is an inherent part of trying to study a dynamic data source. While the database will be updated periodically, it is at present a snapshot in time of the linked web pages, and this is a significant limitation. To remedy this, snapshots of the case study webpages and reports were archived during the data collection process to have a physical record of the source data for this study. These can be made available upon request.

Third, in some business cases the impact claims were not yet verified or audited by a third party, at the time of writing. As a result, due to this self-reporting there is a potential risk of exaggeration of positive impacts. To correct for this limitation, the claims were rigorously checked by the researchers for detailed transparent reporting. Additionally, exact details of the impact claims that qualified cases to be included in the dataset are available in the final Regenerative Business Case Database in Appendix S2, to allow for transparency and verification by the reader.

6. Conclusions

In conclusion, this study conducted a comprehensive practice review of 84 regenerative business cases from 15 sectors. An in-depth analysis of the cases revealed core characteristics related to the size of regenerative organisations, the sectors they operate in, their collaborative networks, how they create and capture value, their ownership structures, the types of regeneration strategies they employ, and how they verify their impact claims. The identified strategies also led to a typology of regenerative business strategies. These fell into six broad archetypal clusters of: nature regeneration, social regeneration, responsible sourcing, human health & well-being focused, employee-level focus. Further, it was also observed that companies often used multiple regenerative strategies concurrently and that partnerships with various stakeholders (such as local community organisations, academic and research partners or indigenous communities) were common. This implies that regeneration is not something an organisation does in isolation, and that regeneration could be most effective through collaboration and being embedded within local contexts.

The regenerative business cases and strategy archetypes identified by this study can serve as a source of inspiration to business designers and entrepreneurs who wish to experiment towards more resilient business models with a stronger sustainability focus. For researchers the database and typology can be a resource for future studies, with the potential for in-depth analysis, case comparison and analysing sectors. Future research avenues could also include longitudinal studies that observe the progress of regenerative companies over time, toto gain more insights on how such innovations occur, succeed (or fail), and create net positive impact (or fail to do so due to rebound effects). This would also help quantify the long-term impact of these businesses on nature and society, which was beyond the scope of this study. There is a need for more action-oriented research aimed at providing support to companies as they transition towards adopting regenerative business models. Future research could also investigate to what extent different regeneration strategies act as direct enablers of regeneration, and which strategies act as more indirect supporters of regeneration. This could be useful in identifying the patterns in which direct and indirect regenerative strategies interact and rely on each other to effect regeneration. These research directions can further our understanding of, and facilitate the practical implementation of regenerative practices within nature and society.

The damage done to the earth by linear and degenerative economies that overexploit natural resources has hampered the natural regenerative capacity of the planet. Going beyond the well-meaning but insufficient practice of circularity, and actively employing regeneration is therefore a necessary part of ensuring the future welfare of the planet and our civilisations. There is also a strong business case for adoption of regenerative business strategies from a resilience, risk management and mitigation perspective, especially in an era of volatile supply chains resulting from climate disasters and associated conflicts. While it might not yet be possible to have a completely regenerative business while existing in a larger linear system, smaller changes can create momentum towards larger systemic shifts.

CRediT authorship contribution statement

Ankita Das: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nancy Bocken:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary data

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References

Albert Heijn, 2024. True-price| Albert Heijn. https://www.ah.nl/over-ah/duurzaamheid/true-price.

Allwood, J. M. (2014). Chapter 30 - squaring the circular Economy: The role of recycling within a hierarchy of material management strategies. In E. Worrell & M. A. Reuter (Eds.), Handbook of Recycling (pp. 445–477). Elsevier. doi:https://doi.org/10.1016/B978-0-12-396459-5.00030-1.

Barredo, L. (2021, July 6). Science Panel for the Amazon: Presentation of Initial Findings. https://www.unsdsn.org/science-panel-for-the-amazon-presentation-of-initial-findings.

Bastounis, A., Buckell, J., Hartmann-Boyce, J., Cook, B., King, S., Potter, C., Bianchi, F., Rayner, M., Jebb, S.A., 2021. The impact of environmental sustainability labels on willingness-to-pay for foods: a systematic review and Meta-analysis of discrete choice experiments. Nutrients 13 (8). https://doi.org/10.3390/nu13082677. Article 8.

Bocken, N., Konietzko, J., 2022. Circular business model innovation in consumer-facing corporations. Technol. Forecast. Soc. Chang. 185, 122076 https://doi.org/10.1016/j.techfore.2022.122076.

Bocken, N., Konietzko, J., 2023. Experimentation capability for a circular economy: a practical guide. J. Bus. Strateg. 44 (6), 406–414. https://doi.org/10.1108/JBS-02-2022-0039.

Bocken, N., Short, S.W., 2021. Unsustainable business models – Recognising and resolving institutionalised social and environmental harm. J. Clean. Prod. 312, 127828 https://doi.org/10.1016/j.jclepro.2021.127828.

Bocken, N., Snihur, Y., 2020. Lean startup and the business model: experimenting for novelty and impact. Long Range Plann. 53 (4), 101953 https://doi.org/10.1016/j.lrp.2019.101953.

Bocken, N., Short, S.W., Rana, P., Evans, S., 2014. A literature and practice review to develop sustainable business model archetypes. J. Clean. Prod. 65, 42–56. https://doi.org/10.1016/j.jclepro.2013.11.039.

Bocken, N., de Pauw, I., Bakker, C., van der Grinten, B., 2016. Product design and business model strategies for a circular economy. J. Ind. Prod. Eng. 33 (5), 308–320. https://doi.org/10.1080/21681015.2016.1172124.

Bocken, N., Niessen, L., Short, S.W., 2022. The sufficiency-based circular Economy—an analysis of 150 companies. Frontiers in Sustainability 3. https://doi.org/10.17863/CAM.84649.

Bosschaert, T. (2022). Circularity Is Not Sustainability: How Well-Intentioned Concepts Distract us from our True Goals, and how SiD Can Help Navigate that Challenge. In *The Impossibilities of the Circular Economy*. Routledge.

Brundtland, G. H. (1987). Our Common Future. Oxford University Press. https://sustainabledevelopment.un.org/content/documents/5987our-common-future.pdf.

Bryman, A., & Bell, E. (2011). *Business Research Methods* (3rd Ed). Oxford University Press.

Bupa. (2024). *Mission Regenerate | Bupa.com*. https://www.bupa.com/sustainability/mission-regenerate.

Caldera, S., Desha, C., Dawes, L., 2018. Exploring the characteristics of sustainable business practice in small and medium-sized enterprises: experiences from the Australian manufacturing industry. J. Clean. Prod. 177, 338–349. https://doi.org/10.1016/j.jclepro.2017.12.265.

Caldera, S., Hayes, S., Dawes, L., Desha, C., 2022. Moving beyond business as usual toward regenerative business practice in small and medium-sized enterprises. Frontiers in Sustainability 3. https://doi.org/10.3389/frsus.2022.799359.

CarbonEthics, 2024. CarbonEthics | Blue Carbon. Carbonethics. https://www.carbonethics.co.

Carlson, B.M., 2011. Principles of Regenerative Biology. Elsevier.

CDP. (2019). Fast Moving Consumers—Which Consumer Goods companies are ready for the low-carbon transition? https://www.cdp.net/en/articles/media/top-fmcgs-in-race-to-keep-up-with-conscious-consumers.

Cetin, S., De Wolf, C., Bocken, N., 2021. Circular digital built environment: an emerging framework. Sustainability 13 (11). https://doi.org/10.3390/su13116348. Article 11.

Channel 4 UK (Director). (2022). Cadbury Exposed: Dispatches. https://www.channel4.com/programmes/cadbury-exposed-dispatches.

ClayTec. (2024). Healthy. ClayTec - Baustoffe aus Lehm. https://claytec.de/en/the-building-material-clay/healthy/.

Community Grains. (2024). Community Grains True Whole Grain—Homepage. https://www.communitygrains.com/, https://www.communitygrains.com/.

Cooperative coffees. (2024). Coop coffees. https://coopcoffees.coop/.

Corbin, J.M., Strauss, A., 1990. Grounded theory research: procedures, canons, and evaluative criteria. Qual. Sociol. 13 (1), 3–21. https://doi.org/10.1007/BF00988593.

B Corp, 2024. What is a B Corp? - B lab UK. https://bcorporation.uk/b-corp-certification/what-is-a-b-corp/.

Crippa, M., Solazzo, E., Guizzardi, D., Monforti-Ferrario, F., Tubiello, F.N., Leip, A., 2021. Food systems are responsible for a third of global anthropogenic GHG emissions. Nature Food 2 (3), 198–209. https://doi.org/10.1038/s43016-021-00225-9.

Das, A., Konietzko, J., Bocken, N., 2022. How do companies measure and forecast environmental impacts when experimenting with circular business models? Sustainable Production and Consumption 29, 273–285. https://doi.org/10.1016/j.spc.2021.10.009.

DaSilva, C.M., Trkman, P., 2014. Business model: what it is and what it is not. Long Range Plann. 47 (6), 379–389. https://doi.org/10.1016/j.lrp.2013.08.004.

Davies, R. (2024, April 19). Unilever to scale back environmental and social pledges. The Guardian. https://www.theguardian.com/business/2024/apr/19/unilever-to-scale-back-environmental-and-social-pledges.

Dean's Beans. (2024). About Dean. Dean's Beans Organic Coffee Company. https://deansbeans.com/pages/about-dean.

Dias, B.D., 2018. Regenerative development – building evolutive capacity for healthy living systems. International Journal of Design & Nature and Ecodynamics 13 (3), 315–323. https://doi.org/10.2495/DNE-V13-N3-315-323.

Dütschke, E., Frondel, M., Schleich, J., Vance, C., 2018. Moral licensing—another source of rebound? *Frontiers in energy research*, *6*. https://www.frontiersin.org/articles/10.3389/fenrg.2018.00038.

Circle Economy, 2024. The Circularity Gap Report 2024 1–44. https://www.circularity-gap.world/2024.

Elf, P., Gatersleben, B., Christie, I., 2019. Facilitating positive spillover effects: new insights from a mixed-methods approach exploring factors enabling people to live more sustainable lifestyles. Front. Psychol. 9 https://doi.org/10.3389/fpsyg.2018.02699.

European Commission, 2023. Proposal for a DIRECTIVE OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on substantiation and communication of explicit environmental claims (green claims Directive). https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52023PC0166.
Fairafric. (2024). *fairafric—Creating jobs, loving chocolate.* https://fairafric.com/en.
Familia Torres, 2024. About Familia Torres | about Familia Torres |. https://www.torres.es/en/about-familia-torres.
FAO. (2011). The State of the World's Land and Water Resources for Food and Agriculture (0 ed.). Routledge. doi:https://doi.org/10.4324/9780203142837.
FCA. (2023). *Policy Statement (PS23/16): Sustainability Disclosure Requirements (SDR) and investment labels.*
Gardner, E.M., Ahmad Puad, A.S., Pereira, J.T., Tagi, J.anak, Nyegang, S. anak, Miun, P., Jumian, J., Pokorny, L., Zerega, N.J.C., 2022. Engagement with indigenous people preserves local knowledge and biodiversity alike. Curr. Biol. 32 (11), R511–R512. https://doi.org/10.1016/j.cub.2022.04.062.
Geissdoerfer, M., Savaget, P., Evans, S., 2017. The Cambridge business model innovation process. Procedia Manufacturing 8, 262–269. https://doi.org/10.1016/j.promfg.2017.02.033.
Ghisellini, P., Cialani, C., Ulgiati, S., 2016. A review on circular economy: the expected transition to a balanced interplay of environmental and economic systems. J. Clean. Prod. 114, 11–32. https://doi.org/10.1016/j.jclepro.2015.09.007.
Gordon, E., Davila, F., Riedy, C., 2023. Regenerative agriculture: a potentially transformative storyline shared by nine discourses. Sustainability Science 18 (4), 1833–1849. https://doi.org/10.1007/s11625-022-01281-1.
Government of UK. (2023, January 26). CMA to Scrutinise 'Green' Claims in Sales of Household Essentials. GOV.UK. https://www.gov.uk/government/news/cma-to-scrutinise-green-claims-in-sales-of-household-essentials.
Greenfield, P., 2023, December 4. UK ad watchdog to crack down on 'biodegradable' and 'recyclable' claims. The Guardian. https://www.theguardian.com/environment/2023/dec/04/ad-regulator-to-crack-down-on-biodegradable-and-compostable-claims-aoe.
Greyston Bakery, 2024. Greyston Bakery | learn about Greyston Bakery. Greyston. http://www.greyston.org/?who_we_are.
Griese, K.-M., Werner, K., Hogg, J., 2017. Avoiding greenwashing in event marketing: an exploration of concepts, literature and methods. Journal of Management and Sustainability 7 (4), 1–15.
Gualandris, J., Branzei, O., Wilhelm, M., Lazzarini, S., Linnenluecke, M., Hamann, R., Dooley, K.J., Barnett, M.L., Chen, C.-M., 2024. Unchaining supply chains: transformative leaps toward regenerating social–ecological systems. J. Supply Chain Manag. 60 (1), 53–67. https://doi.org/10.1111/jscm.12314.
Guayaki, 2023. Regeneration. Guayakí Yerba Mate. https://shop.guayaki.com/pages/regeneration.
Hahn, T., Tampe, M., 2021. Strategies for regenerative business. Strateg. Organ. 19 (3), 456–477. https://doi.org/10.1177/1476127020979228.
Hardman, G.J., 2009. *Regenerative leadership: an integral theory for transforming people and organizations for sustainability in business, education, and community* [doctoral thesis, Florida Atlantic University]. https://www.academia.edu/306961/Regenerative_Leadership_An_Integral_Theory_for_Transforming_People_and_Organizations_for_Sustainability_in_Business_Education_and_Community.
Hardman, G. J. (2013). Leading for Regeneration: Going beyond Sustainability in Business Education, and Community. Routledge.
Hawken, P. (2021). Regeneration: Ending the Climate Crisis in One Generation. Penguin.
Herrmann, C., Rhein, S., Sträter, K.F., 2022. Consumers' sustainability-related perception of and willingness-to-pay for food packaging alternatives. Resources, Conservation and Recycling 181, 106219. https://doi.org/10.1016/j.resconrec.2022.106219.
van Hille, I., de Bakker, F.G.A., Groenewegen, P., Ferguson, J.E., 2021. Strategizing nature in cross-sector partnerships: can plantation revitalization enable living wages? Organ. Environ. 34 (2), 175–197. https://doi.org/10.1177/1086026619886848.
Hockerts, K., Wüstenhagen, R., 2010. Greening goliaths versus emerging Davids—theorizing about the role of incumbents and new entrants in sustainable entrepreneurship. J. Bus. Ventur. 25 (5), 481–492. https://doi.org/10.1016/j.jbusvent.2009.07.005.
Hoveskog, M., Halila, F., Mattsson, M., Upward, A., Karlsson, N., 2018. Education for sustainable development: business modelling for flourishing. J. Clean. Prod. 172, 4383–4396. https://doi.org/10.1016/j.jclepro.2017.04.112.
Howard, M., Hopkinson, P., Miemczyk, J., 2019. The regenerative supply chain: a framework for developing circular economy indicators. Int. J. Prod. Res. 57 (23), 7300–7318. https://doi.org/10.1080/00207543.2018.1524166.
IKEA. (2024). People & Planet – IKEA Global. IKEA. https://www.ikea.com/global/en/our-business/people-planet/.
Ioannou, I., Kassinis, G., Papagiannakis, G., 2022, July 21. How greenwashing affects the bottom line. Harv. Bus. Rev. https://hbr.org/2022/07/how-greenwashing-affects-the-bottom-line.
IPCC. (2022). Climate change 2022 – Impacts, adaptation and vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (1st ed.). Cambridge University Press. doi:https://doi.org/10.1017/9781009325844.
Jaeger-Erben, M., Jensen, C., Hofmann, F., Zwiers, J., 2021. There is no sustainable circular economy without a circular society. Resour. Conserv. Recycl. 168, 105476 https://doi.org/10.1016/j.resconrec.2021.105476.
Kaddoura, M., Kambanou, M.L., Tillman, A.-M., Sakao, T., 2019. Is prolonging the lifetime of passive durable products a low-hanging fruit of a circular Economy? A multiple case study. Sustainability 11 (18). https://doi.org/10.3390/su11184819. Article 18.
Konietzko, J., Bocken, N., Hultink, E.J., 2020. A tool to analyze, ideate and develop circular innovation ecosystems. Sustainability 12 (1), 417. https://doi.org/10.3390/su12010417.
Konietzko, J., Das, A., Bocken, N., 2023. Towards regenerative business models: a necessary shift? Sustainable Production and Consumption 38, 372–388. https://doi.org/10.1016/j.spc.2023.04.014.
Leal Filho, W., Perry, P., Heim, H., Dinis, M.A.P., Moda, H., Ebhuoma, E., Paço, A., 2022. An overview of the contribution of the textiles sector to climate change. Front. Environ. Sci. 10 https://doi.org/10.3389/fenvs.2022.973102.
Liu, A. (2024, January 4). To thrive businesses must set their sights on regeneration. World Economic Forum. https://www.weforum.org/agenda/2024/01/business-resilience-regeneration/.
Lozano-García, B., Francaviglia, R., Renzi, G., Doro, L., Ledda, L., Benítez, C., González-Rosado, M., Parras-Alcántara, L., 2020. Land use change effects on soil organic carbon store. An opportunity to soils regeneration in Mediterranean areas: implications in the 4p1000 notion. Ecol. Indic. 119, 106831 https://doi.org/10.1016/j.ecolind.2020.106831.
Lüdeke-Freund, F., Gold, S., Bocken, N., 2019. A review and typology of circular Economy business model patterns. J. Ind. Ecol. 23 (1), 36–61. https://doi.org/10.1111/jiec.12763.
LUSH, 2024a. Leaving the world lusher than we found it. We Are Lush. https://weare.lush.com/lush-life/our-impact-reports/leaving-the-world-lusher-than-we-found-it/.
LUSH. (2024b). Our Values. We Are Lush. https://weare.lush.com/lush-life/our-values/.
MacDonald, C. (2018, May 31). How Coke spun the public on its water use. The Verge. https://www.theverge.com/2018/5/31/17377964/coca-cola-water-sustainability-recycling-controversy-investigation.
Michalke, A., Stein, L., Fichtner, R., Gaugler, T., Stoll-Kleemann, S., 2022. True cost accounting in Agri-food networks: a German case study on informational campaigning and responsible implementation. Sustain. Sci. 17 (6), 2269–2285. https://doi.org/10.1007/s11625-022-01105-2.
Mondelez International, 2024. Cocoa life—about the program. Cocoa Life. https://www.cocoalife.org/the-program/approach/.
Montgomery, A.W., Lyon, T.P., Barg, J., 2023. No end in sight? A greenwash review and research agenda. Organ. Environ., 10860266231168905 https://doi.org/10.1177/10860266231168905.
Moreno, M., De los Rios, C., Rowe, Z., Charnley, F., 2016. A conceptual framework for circular design. Sustainability 8 (9). https://doi.org/10.3390/su8090937. Article 9.
Morseletto, P., 2020. Restorative and regenerative: exploring the concepts in the circular economy. Journal of Industrial Ecology 24 (4), 763–773. https://doi.org/10.1111/jiec.12987.
Muñoz, P., Branzei, O., 2021. Regenerative organizations: introduction to the special issue. Organ. Environ. 34 (4), 507–516. https://doi.org/10.1177/10860266211055740.
Native. (2024). Native—Produtos Orgânicos—Página Inicial. Native - Produtos Orgânicos - Página Inicial. https://www.nativealimentos.com.br/en.
Nemes, N., Scanlan, S.J., Smith, P., Smith, T., Aronczyk, M., Hill, S., Lewis, S.L., Montgomery, A.W., Tubiello, F.N., Stabinsky, D., 2022. An integrated framework to assess greenwashing. Sustainability 14 (8). https://doi.org/10.3390/su14084431. Article 8.
Norris, G. A. (2015). *Handprint-Based Netpositive Assessment.* (Sustainability and Health Initiative for NetPositive Enterprise (SHINE)). Center for Health and the Global Environment, Harvard T. H. Chan School of Public Health.
Norris, G.A., Burek, J., Moore, E.A., Kirchain, R.E., Gregory, J., 2021. Sustainability health initiative for NetPositive Enterprise handprint methodological framework. Int. J. Life Cycle Assess. 26 (3), 528–542. https://doi.org/10.1007/s11367-021-01874-5.
Persson, L., Carney Almroth, B.M., Collins, C.D., Cornell, S., de Wit, C.A., Diamond, M.L., Fantke, P., Hassellöv, M., MacLeod, M., Ryberg, M.W., Søgaard Jørgensen, P., Villarrubia-Gómez, P., Wang, Z., Hauschild, M.Z., 2022. Outside the safe operating space of the planetary boundary for novel entities. Environ. Sci. Technol. 56 (3), 1510–1521. https://doi.org/10.1021/acs.est.1c04158.
True Pigments, 2024. True Pigments | quality paints from pollution | cleaning Appalachian streams of Iron oxide. https://www.truepigments.com//.
Polman, P., Winston, A.S., 2021. Net Positive: How Courageous Companies Thrive by Giving More than they Take. Harvard Business Review Press.
Quarshie, A., Salmi, A., Wu, Z., 2021. From equivocality to reflexivity in biodiversity protection. Organ. Environ. 34 (4), 530–558. https://doi.org/10.1177/1086026619837122.
Rhodes, C.J., 2015. Permaculture: regenerative – not merely sustainable. Sci. Prog. 98 (4), 403–412. https://doi.org/10.3184/003685015X14467291596242.
Richardson, K., Steffen, W., Lucht, W., Bendtsen, J., Cornell, S.E., Donges, J.F., Drüke, M., Fetzer, I., Bala, G., von Bloh, W., Feulner, G., Fiedler, S., Gerten, D., Gleeson, T., Hofmann, M., Huiskamp, W., Kummu, M., Mohan, C., Nogués-Bravo, D., Rockström, J., 2023. Earth beyond six of nine planetary boundaries. *Science*. Advances 9 (37), eadh2458. https://doi.org/10.1126/sciadv.adh2458.
Ricoh. (2024). Conservation of Biodiversity | Global | Ricoh. https://www.ricoh.com/sustainability/environment/biodiversity.
Ritala, P., Huotari, P., Bocken, N., Albareda, L., Puumalainen, K., 2018. Sustainable business model adoption among S&P 500 firms: a longitudinal content analysis study. J. Clean. Prod. 170, 216–226. https://doi.org/10.1016/j.jclepro.2017.09.159.
Robertson, J.L., Montgomery, A.W., Ozbilir, T., 2023. Employees' response to corporate greenwashing. Bus. Strateg. Environ. 32 (7), 4015–4027. https://doi.org/10.1002/bse.3351.
Robinson, J., Cole, R.J., 2015. Theoretical underpinnings of regenerative sustainability. Building Research & Information 43 (2), 133–143. https://doi.org/10.1080/09613218.2014.979082.

Roland, E., & Landua, G. (2015). Regenerative enterprise: Ptimizing for multi-capital abundance. Lulu Press, Inc.
Sanders, A., 2022. Binding capital to free purpose: steward ownership in Germany. European Company and Financial Law Review 19 (4), 622–653. https://doi.org/10.1515/ecfr-2022-0020.
Sanford, C., 2017. The Regenerative Business: Redesign Work, Cultivate Human Potential, Achieve Extraordinary Outcomes (1st edition). Nicholas Brealey Publishing.
Saunders, M., Lewis, P., Thornhill, A., 2009. Research Methods for Business Students (5. ed). Financial Times Prentice Hall.
Serendipalm, 2024. Organic Fair Trade Palm Oil. Serendipalm. https://serendipalm.com/.
Slawinski, N., Winsor, B., Mazutis, D., Schouten, J.W., Smith, W.K., 2021. Managing the paradoxes of place to Foster regeneration. Organ. Environ. 34 (4), 595–618. https://doi.org/10.1177/1086026619837131.
Stinner, D.H., Stinner, B.R., Martsolf, E., 1997. Biodiversity as an organizing principle in agroecosystem management: case studies of holistic resource management practitioners in the USA. Agr Ecosyst Environ 62 (2), 199–213. https://doi.org/10.1016/S0167-8809(96)01135-8.
Stubbs, W., & Cocklin, C. (2008). Conceptualizing a "Sustainability Business Model." Organ. Environ., 21(2), 103–127. doi:https://doi.org/10.1177/1086026608318042.
Ungoed-Thomas, J., 2022, April 3. Cadbury faces fresh accusations of child labour on cocoa farms in Ghana. The Observer. https://www.theguardian.com/law/2022/apr/03/cadbury-faces-fresh-accusations-of-child-labour-on-cocoa-farms-in-ghana.
Vanham, 2024. How America's sustainability backlash is causing a transatlantic divide. Fortune. https://fortune.com/2024/03/28/how-americas-sustainability-backlash-is-causing-a-transatlantic-divide/.
Visser, W., Crane, A., 2010. Corporate sustainability and the individual: understanding what drives sustainability professionals as change agents. SSRN Electron. J. https://doi.org/10.2139/ssrn.1559087.
Playa Viva, 2023. Playa Viva Is more than Just a Hotel. We're Commited to Making an Impact, Playa Viva. https://www.playaviva.com/impact.
Vlasov, M., 2021. In transition toward the Ecocentric entrepreneurship Nexus: how nature helps entrepreneurs make ventures more regenerative over time. Organ. Environ. 34 (4), 559–580. https://doi.org/10.1177/1086026619831448.
WWF. (2022). Living Planet Report 2022—Building a Nature-Positive Society. WWF. https://livingplanet.panda.org/en-US/.
Young, T.P., 2000. Restoration ecology and conservation biology. Biol. Conserv. 92 (1), 73–83. https://doi.org/10.1016/S0006-3207(99)00057-9.
Zermeño-Hernández, I., Pingarroni, A., Martínez-Ramos, M., 2016. Agricultural land-use diversity and forest regeneration potential in human- modified tropical landscapes. Agr Ecosyst Environ 230, 210–220. https://doi.org/10.1016/j.agee.2016.06.007.
Zink, T., Geyer, R., 2017. Circular Economy rebound. Journal of Industrial Ecology 21 (3), 593–602. https://doi.org/10.1111/jiec.12545.